

#3.5.6.

Let  $f: [a, b] \rightarrow \mathbb{R}$  be a continuous map.

Show that  $g(x) = \max \{f(y) \mid a \leq y \leq x\}$ ,  $x \in [a, b]$  is continuous.

Proof. Given  $x_0 \in [a, b]$  and arbitrary  $\epsilon > 0$ , there is  $\delta > 0$  such that  
 $|x_0 - t| < \delta \Rightarrow |f(x_0) - f(t)| < \epsilon$ .

Meanwhile, observe that for  $x \leq t$ ,

$$\max \{f(y) \mid a \leq y \leq t\} = \max \left\{ \max \{f(y) \mid a \leq y \leq t\}, \max \{f(y) \mid t \leq y \leq x\} \right\}$$

#3.5.8. Let  $f: (0, \infty) \rightarrow \mathbb{R}$  be continuous and  $\lim_{x \rightarrow \infty} [f(x+1) - f(x)] = \alpha$ . Then,  

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x} = \alpha.$$

Proof. Observe that: given  $N \in \mathbb{N}$ ,

$$f(x+N) - f(x) = \sum_{i=0}^{N-1} f(x+(i+1)) - f(x+i)$$

First, assume that  $\alpha = 0$ . Then, given  $\epsilon > 0$ , there exists  $\delta > 0$  s.t.

$$x \geq \delta \Rightarrow |f(x+1) - f(x)| < \epsilon.$$

Since  $f$  is continuous,  $f$  is bounded on  $[\delta, \delta+1]$ . Let  $M > 0$  s.t.  $|f(x)| \leq M$ ,  $x \in [\delta, \delta+1]$ .

Now, for  $t \in [\delta, \delta+1]$  and arbitrary  $N \in \mathbb{N}$ ,

$$\begin{aligned} |f(t+N) - f(t)| &= \left| \sum_{i=0}^{N-1} f(t+(i+1)) - f(t+i) \right| \\ &\leq \sum_{i=0}^{N-1} |f(t+(i+1)) - f(t+i)| \\ &\leq N \cdot \epsilon \end{aligned}$$

So,  $|f(t+N)| \leq N\epsilon + |f(t)| \leq N\epsilon + M$ , for all  $t \in [\delta, \delta+1]$ .

Now, for any  $x > \delta$ ,

$$|f(x)| \leq \lceil x - \delta \rceil \cdot \epsilon + M, \text{ so}$$

$$\left| \frac{f(x)}{x} \right| \leq \frac{\lceil x - \delta \rceil \cdot \epsilon}{x} + \frac{M}{x} \rightarrow 0 \text{ as } x \rightarrow \infty.$$

Now, for arbitrary  $\alpha \in \mathbb{R}$ , let  $g(x) = f(x) - \alpha x$ .

73.5.12.

Let  $f: [0, \infty) \rightarrow \mathbb{R}$  be uniformly continuous s.t.

$$\lim_{n \rightarrow \infty} f(n+x) = 0, \quad x \in [0, 1].$$

Then  $\lim_{x \rightarrow \infty} f(x) = 0$ .

Proof. Given  $\epsilon > 0$ , there is  $N \in \mathbb{N}$  s.t.  $|x-y| \leq \frac{1}{N} \Rightarrow |f(x) - f(y)| < \epsilon$

Now, for each  $0, \frac{1}{N}, \frac{2}{N}, \dots, \frac{N-1}{N}, 1$ , there is  $r_j \in \mathbb{N}$  s.t.

$$n \geq r_j \Rightarrow |f(n + \frac{j}{N})| < \epsilon$$

Now, put  $K = \max\{r_0, \dots, r_N\}$ ,

for any  $x \geq K$ ,  $x \in [p + \frac{j-1}{N}, p + \frac{j}{N}]$  for some  $p \geq K$  and  $p \in \mathbb{N}$ ,  $j \in \{1, \dots, N\}$ .

$$\text{So, } |f(x)| = |f(x) - f(p + \frac{j}{N}) + f(p + \frac{j}{N})| \leq \epsilon + \epsilon = 2\epsilon$$

73.5.17. Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be a map satisfying

$$e^x f(x) \quad \text{and} \quad e^{-x} f(x)$$

both are monotone decreasing. Prove that  $f$  is continuous.

Proof.

14.4.7,

Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be a differentiable map.

If  $f'$  is bounded, then  $f$  is continuous uniformly.

Proof. Let  $M > 0$  be a bound for  $f'$ . There is,  $|f'(c)| < M$  for all  $x$ .

Given  $x, y \in \mathbb{R}$ ,  $|f(y) - f(x)| = |f'(c)| \cdot |x - y| \leq M \cdot |x - y|$  for some  $c$  between  $x$  and  $y$ .

Let  $\varepsilon > 0$  be given. Then, for  $\delta = \frac{\varepsilon}{M}$ ,  $|x - y| < \delta$  implies

$$|f(x) - f(y)| \leq M \cdot |x - y| < \varepsilon.$$

14.4.8

Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be a map satisfying

$$|f(x) - f(y)| \leq |x - y|^2 \text{ for all } x, y \in \mathbb{R},$$

prove that  $f$  is constant.

Proof. Given  $x_0 \in \mathbb{R}$  and  $1 > \varepsilon > 0$ ,

$$0 < |x_0 - x| < \varepsilon \text{ implies } \frac{|f(x_0) - f(x)|}{|x_0 - x|} \leq |x_0 - x| < \varepsilon$$

so,  $f$  is differentiable, and  $f'(x_0) = 0$ .

Since  $x_0$  was arbitrary,  $f' = 0$ , so  $f$  is constant.

14.4.a.

Let  $f: (a, b) \rightarrow \mathbb{R}$  be a continuous on  $(a, b)$  and differentiable on  $(a, b) \setminus \{c\}$  where  $a < c < b$ .

If  $\lim_{x \rightarrow c} f'(x)$  exists, then  $f$  is differentiable at  $c$  with  $f'(c) = \lim_{x \rightarrow c} f'(x)$ .

Proof. Given  $\varepsilon > 0$ , there is  $\delta > 0$  s.t.  $0 < |x - c| < \delta \Rightarrow |L - f'(x)| < \varepsilon$ , where  $L = \lim_{x \rightarrow c} f'(x)$ .

Let  $t \in (c, c + \delta)$  be given. Then, for some  $c' \in (c, t)$ ,

$$\frac{f(t) - f(c)}{t - c} = f'(c')$$

And, similarly, for  $t \in (c - \delta, c)$ , there is  $c'' \in (t, c)$  s.t.

$$\frac{f(t) - f(c)}{t - c} = f'(c'').$$

by the Mean-Value Theorem. Now, for any  $0 < |t - c| < \delta$ ,

$$\left| \frac{f(t) - f(c)}{t - c} - L \right| = |f'(c') - L| < \varepsilon, \text{ for some } c' \in (c - \delta, c + \delta),$$

14.4.1. Let  $f: I \rightarrow \mathbb{R}$  be a differential function on an interval  $I$ .

If  $f'(x) \neq \alpha$  for all  $x \in I$ , then  $f' > \alpha$  or  $f' < \alpha$ .

proof

[Integral

#5, 6, 3,

If  $f: [a, b] \rightarrow \mathbb{R}$  is Riemann-integrable, then so is  $f^2$  and  $|f|$ .

And deduce that if  $f$  and  $g$  are integrable, so is  $fg$ .

Proof. Given  $\epsilon > 0$ , there is a partition  $P = \{x_0, \dots, x_n\}$  on  $[a, b]$  such that

$$U(f, P) - L(f, P) = \sum_{i=1}^n (M_i - m_i) \cdot \Delta_i$$

where  $\Delta_i = |x_i - x_{i-1}|$ ,  $M_i = \sup \{f(x) \mid x \in [x_{i-1}, x_i]\}$ ,  $m_i = \inf \{f(x) \mid x \in [x_{i-1}, x_i]\}$ .

For any  $x, y$ ,  $|f(x) - f(y)| \leq |f(x) + f(y)|$ , therefore for  $I_i = [x_{i-1}, x_i]$ ,

$$\sup_{I_i} |f| - \inf_{I_i} |f| \leq \sup_{I_i} f - \inf_{I_i} f,$$

therefore  $U(|f|, P) - L(|f|, P) < \epsilon$  so  $|f|$  is Riemann-integrable.

Now, let  $M > 0$  be a bound for  $|f|$  on  $[a, b]$ .

$$\text{Given } x, y, |f(x)^2 - f(y)^2| = |(f(x) + f(y))(f(x) - f(y))| \leq 2M \cdot |f(x) - f(y)|.$$

Given  $\epsilon > 0$ , take  $P = \{x_0, \dots, x_n\}$  s.t.

$$U(f, P) - L(f, P) < \frac{\epsilon}{2M}.$$

$$\text{Now, } U(f^2, P) - L(f^2, P) = \sum_{i=1}^n (\sup_{I_i} f^2 - \inf_{I_i} f^2) \cdot \Delta_i$$

$$\leq \sum_{i=1}^n 2M (\sup_{I_i} f - \inf_{I_i} f) \cdot \Delta_i$$

$$< \epsilon. \quad \text{So proved.}$$

Now, by the equation

$$fg = \frac{1}{4}((f+g)^2 - (f-g)^2)$$

gives the result.

15.6.5.

Let  $f: [0,1] \rightarrow \mathbb{R}$  be a continuous map such that  $f \geq 0$  and  $\int_0^1 f = 0$ .

Then  $f \equiv 0$ .

Proof. Suppose that  $f \not\equiv 0$ . Choose  $t_0 \in [0,1]$  s.t.  $f(t_0) > 0$ .

Since  $f$  is continuous, there is  $\delta > 0$  such that

$$|x - t| < \delta \Rightarrow |f(x) - f(t)| < \frac{f(t)}{2}$$

And, by the triangle inequality,

$$-\frac{f(t)}{2} < |f(x)| - |f(t)| < \frac{f(t)}{2}$$

$$\text{or } \left| \frac{f(x)}{2} \right| < |f(x)| < \left| \frac{3f(x)}{2} \right| \text{ if } x \in (t - \delta, t + \delta).$$

Now, for any partition  $P$  on  $[a,b]$  the refinement

$$P^* = P \cup \{[a,b] \cap \{t - \delta, t + \delta\}\}$$

$$\text{satisfies } \mathcal{U}(f, P) \geq \mathcal{U}(f, P^*) \geq 2\delta \cdot \frac{f(t)}{2} = \delta f(t)$$

But, since  $\int_0^1 f = 0$ , we can take  $P'$  s.t.  $\mathcal{U}(f, P') < \delta f(t)$ , it is contradiction.

15.6.6.

Let  $f: [a, b] \rightarrow \mathbb{R}$  be a continuous map. Then,

$$\lim_{n \rightarrow \infty} \left( \int_a^b |f(x)|^n dx \right)^{1/n} = \max \{ |f(x)| \mid x \in [a, b] \}$$

Proof. Let  $M = \max \{ |f(x)| \mid x \in [a, b] \}$ . If  $M=0$ , then  $f=0$ , so there's nothing to prove.

Assume  $M > 0$ .



15.6.10.

Let  $f, g: [a, b] \rightarrow \mathbb{R}$  be continuous maps, and  $g \geq 0$ .

Then, for some  $c \in [a, b]$ ,  $\int_a^b f(x)g(x) dx = f(c) \int_a^b g(x) dx$ .

Proof. WLOG, let  $a=0$  and  $b=1$ . And, if  $\int_0^1 g = 0$ , then  $g=0$  since  $g \geq 0$ . Let  $\int_0^1 g \neq 0$ .  
Let  $m = \min\{f(x) | x \in [0, 1]\}$  and  $M = \max\{f(x) | x \in [0, 1]\}$ . Then,

$mg \leq fg \leq Mg$  for all  $x \in [0, 1]$ , so therefore

$$m \cdot \int_0^1 g \leq \int_0^1 fg \leq M \cdot \int_0^1 g.$$

$$\Rightarrow m \leq \frac{\int_0^1 fg}{\int_0^1 g} \leq M.$$

Now by the IVT,  $f(c) = \frac{\int_0^1 fg}{\int_0^1 g}$  for some  $c$ .

15.6.11. Bounded Variation always has the side limits

Proof. Let  $f: [a, b] \rightarrow \mathbb{R}$  be the bounded variation.

Let  $x \in (a, b)$  be given.

